
HOMEWORK DAY 39 – *Separable Equations §9.3*

1. Solve the differential equation or initial value problem.

(a) $\frac{dy}{dx} = 3x^2y^2$

(b) $\frac{dy}{dx} = x\sqrt{y}$

(c) $xy' = 2y + 1, \quad y(1) = 0$

(d) $\frac{dy}{dx} = 2x(y^2 + 1)$

(e) $\frac{dp}{dt} = t^2 p - p + t^2 - 1$

HOMEWORK DAY 40 – *Integrating factors (See posted example) §9.5*

2. Which of the equations in §9.5:1-4 is linear?
3. Find a positive solution $\mu(x)$ to each of the following equations. Except for (a,b), you need to show all your work.

(a) $\mu' = -\mu$

(b) $\mu' = 2\mu$

(c) $\mu' = \frac{\mu}{x}$

(d) $\mu' = \frac{4\mu}{x}$

4. For each of the following first order linear equations, following the “Worked out Example” on the course website,
- (i) Place the equation in the form $y' + p(t)y = q(t)$.
 - (ii) Multiply the equation by μ .
 - (iii) State the equation that μ must satisfy so that the left hand side of (ii) is an exact derivative.
 - (iv) Find a positive solution μ to the equation in (iii), using separation of variables if needed.
 - (v) Proceed to solve the equation in (ii).
- (a) $4x^3y + x^4y' = \sin^3 x$

(b) $t^2 \frac{dy}{dt} + 3ty = \sqrt{1+t^2}, t > 0$

5. Solve $y' = y - 2x$, $y(1) = 0$, showing all your work, as above.

6. Show that the following terms are exact derivatives (they are of the form $\frac{d}{dx}[g(x)y]$ for some g).

(a) $xy' + y$

(b) $t^3 \frac{dy}{dt} + 3t^2 y$

7. Use your answer in (a) above to solve $xy' + y = \sqrt{x}$